

Artifact removal improves electrodermal waveforms but not downstream classification in a virtual-reality balance task

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Abstract

Artifact removal is a routine step before electrodermal activity (EDA) is used for classification, on the assumption that a cleaner signal supports a better decision. We tested this assumption in a virtual-reality (VR) balance-disturbance task. A residual gating network was trained on a benchmark with expert-corrected EDA, frozen, and applied to VR recordings, where raw and processed signals were compared under identical leave-one-participant-out evaluation with five published time-series classifiers. On the benchmark the gate detected artifacts well (median record AUROC 0.94) and reduced error inside artifact regions by 17.8%. In the VR task it did not improve classification. Changes in balanced accuracy ranged from -1.35 to $+0.93$ percentage points, no classifier improved significantly, two were worse, and all five were equivalent to raw input within ± 3.32 points. Three observations explain the gap. The correction that lowered waveform error also reduced skin conductance response detection in all 43 benchmark records. Processing left 92.8% of predictions unchanged, and the predictions it did change were corrected and corrupted at similar rates. The VR recordings carried little contamination (an estimated 4.6% of samples), and even perfect localization of deliberately injected artifacts recovered only 3.3 points in the most sensitive classifier. A pooled association between artifact level and accuracy (11.3 points) disappeared within participants (0.1 points), which shows how differences between people can make cleaning appear useful. Preprocessing should be judged by the decision it supports, and an unprocessed arm should be kept for that comparison. **Keywords:** electrodermal activity; artifact removal; virtual reality; equivalence testing; preprocessing evaluation; negative result

1 Introduction

Physiological sensing can tell an immersive virtual-reality (VR) system how its user is responding without interrupting the interaction. Electrodermal activity (EDA) is a natural candidate. It is inexpensive to record with wearable sensors and tracks sympathetic arousal through tonic skin conductance and transient skin conductance responses (SCRs) [1–3]. It is also easily disturbed by movement, because pressure and shifts at the electrode produce deflections that resemble responses [4, 5]. Most applied pipelines therefore remove artifacts before inference. The question we ask is whether that step improves the decision the system is built to make.

The answer is not obvious. A balance disturbance changes movement and physiology at the same moment, so a correction that removes motion-related variation can also remove information that separates one user state from another. More generally, an artifact handler is trained and scored on how closely it reproduces a clean waveform, while a classifier uses whatever features separate its classes. These two targets need not agree.

Processing choices are known to change EDA results. Multiverse analyses of SCR quantification give different effect estimates under different pipelines [6, 7], filtering changes sensitivity [8], and the choice of model inversion changes predictive validity [9, 10]. A few studies have compared processed with unprocessed signals on the same task, one reporting a benefit [11] and another a small loss [12]. In EEG, automated cleaning has rarely increased condition effects [13], and component-based artifact rejection did not improve deep-network decoding in three brain-computer interface tasks [14]. In applied EDA work such direct comparisons remain uncommon. Of 13 applied studies in a documented sample that removed EDA artifacts before a downstream task, only 2 compared processed and unprocessed signals with the same model and protocol (section 2.7). Where the comparison is missing, the benefit of cleaning is assumed rather than shown.

We therefore tested artifact removal against the task it is meant to serve. A continuous residual gate was trained on a public benchmark with expert-corrected EDA [15, 16], frozen, and transferred to a VR balance-disturbance corpus [17], where raw and gated signals were classified by five published time-series methods under identical participant-disjoint folds. The gate performed well on the benchmark but did not improve classification in VR. Because the benchmark provides expert references and the VR corpus provides task labels, we could follow the expected benefit from contamination to decision and locate where it was lost: in event preservation, in the direction of feature changes, in decisions that did not change, and in the small amount of contamination available to remove. We also show that a pooled association between artifact level and accuracy, the kind of evidence often used to justify cleaning, largely reflects differences between participants.

2 Methods

2.1 Data

Two public corpora were used, each for what it can support (figure 1, table 1). EDABE v2 [15, 16] contains 43 EDA records sampled at 128 Hz, each with an expert-corrected reference waveform and a point-wise artifact mask. Annotated artifacts cover 10.75% of all samples, and a fixed split assigns 33 records to training and 10 to testing. This corpus allows supervised training and a direct check of detection, reconstruction and event preservation, but it has no VR task.

The VR balance-disturbance corpus [17] provides task labels but no clean reference. After removal of one duplicated file (P11/P12), it contains 11 participants. Recordings include Xsens motion capture, EMG and Shimmer EDA; we used EDA at 58.8 Hz.

2.2 Task

Each disturbance onset defined a pair of windows: a positive window from 1 to 13 s after onset and a negative window

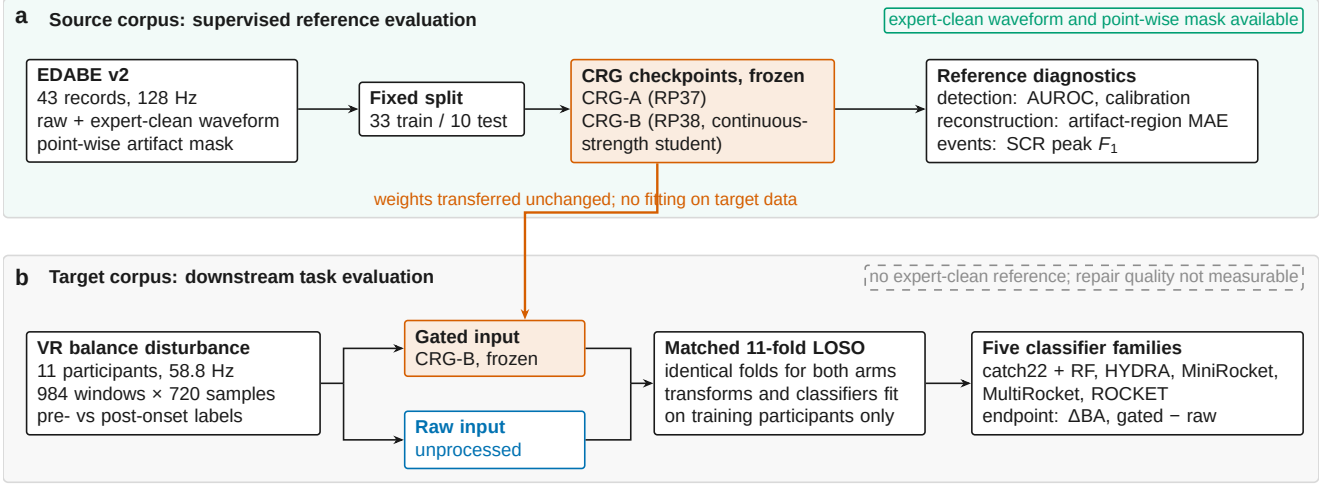


Figure 1: Study design. (a) The benchmark (EDABE v2) supplies expert-clean waveforms and point-wise artifact masks, so detection, reconstruction and event preservation can be scored against a reference. Two checkpoints, CRG-A and CRG-B, were trained on its fixed record split and then frozen. (b) The VR balance-disturbance corpus supplies task labels but no clean reference. The frozen CRG-B operator is applied without fitting on VR data, and raw and gated inputs pass through identical leave-one-participant-out folds, with transforms and classifiers fitted on training participants only. Seeds 1005, 2004 and 2027 are averaged within each of 20,000 participant-cluster bootstrap resamples.

Table 1: Roles of the two corpora. Only the benchmark supports expert-referenced evaluation of the artifact handler; only the VR corpus supplies task labels.

Property	EDABE v2 (benchmark)	VR balance disturbance
Unit	43 records	11 participants
Sampling rate	128 Hz	58.8 Hz
Reference	Expert mask and clean waveform	None
Evaluation	Fixed split, 33 training / 10 test records	11-fold leave-one-participant-out
Task windows	None	984 windows of 720 samples
Artifact burden	10.75% annotated	4.64% estimated

from 22 to 10 s before it. A pair was discarded if either window crossed a trajectory boundary or overlapped another event, which kept the classes balanced within each participant. These rules were fixed before the labels were accessed. The final set contains 984 windows of 720 samples (12.24 s at 58.8 Hz). Because the positive window follows onset, the task separates the states before and after a disturbance; it does not test prediction of a future loss of balance.

2.3 Artifact handler

The continuous residual gate (CRG; figure 2) corrects a trace by adding a learned residual weighted by a learned artifact probability:

$$\begin{aligned}\tilde{x}_{\text{gate}}(t) &= \max\{\tilde{x}(t) + p(t)r(t), 0\}, \\ p(t) &= \sigma(\ell(t)), \quad r(t) = 4 \tanh(d(t)),\end{aligned}\quad (1)$$

where $\tilde{x}(t) = x(t)/s$ is the raw trace divided by a robust per-record scale,

$$s = \max\{P_{95}(x) - P_5(x), 0.05 \mu\text{S}\}, \quad (2)$$

and $\ell(t)$ and $d(t)$ are the outputs of a detection head and a residual head. Both heads share a temporal convolutional trunk with 48 channels and nine residual blocks with dilations from 1 to 256, which gives a receptive field of 4089 samples (about 32 s at 128 Hz). The corrected trace is multiplied by s to return to microsiemens. Where $p(t)$ is near zero the trace passes unchanged, and the clamp keeps conductance non-negative.

Training combined a class-balanced cross-entropy for detection, an L_1 loss to the expert reference inside artifact regions,

and an L_1 loss to the raw signal outside them; the last term discourages changes where none are needed. Two checkpoints were trained with this architecture on the same record split. CRG-A used the three losses with equal weights. CRG-B first trained a fold-local teacher for 12 epochs and then a student for 18 epochs whose objective added 0.15 times a binary cross-entropy to the detached teacher probabilities. All weights were frozen before any downstream analysis.

The two checkpoints serve different analyses. Detection, the waveform examples and the feature analysis use CRG-A; benchmark reconstruction and event analyses use the continuous-strength CRG-B student; the VR task uses CRG-B; and the injection experiment applies CRG-A window by window. The evidence across analyses therefore concerns one method rather than one identical output followed from start to finish, although reconstruction and event preservation are compared within the same checkpoint and records. Archive identifiers for each checkpoint are listed in the supplementary package.

Event preservation was measured on the benchmark. SCR peaks were detected on each signal after a second-order 0.05 Hz high-pass filter (prominence at least 0.01 μS , minimum separation 1 s) and matched one-to-one to peaks of the expert reference within ± 1 s. Peak F_1 summarizes this matching.

2.4 Downstream evaluation

Five published classifier families were used without modification: catch22 features with a random forest [18], ROCKET [19], MiniRocket [20], MultiRocket [21] and HYDRA [22]. They range from 22 interpretable features to large banks of random convolutional kernels, so a result that holds across them does not depend on one way of building features. The families were fixed before any treatment effect was seen.

Evaluation used 11-fold leave-one-participant-out cross-validation with seeds 1005, 2004 and 2027. Raw and gated arms used the same folds. Within each fold and seed, kernel transforms were fitted on training participants only and shared across arms, and each arm fitted its own scaler and classifier. The primary endpoint was the difference in balanced accuracy (Δ BA), gated minus raw, in percentage points (pp). The decision analysis in section 3.5 uses pooled accuracy instead, because pooled accuracy decomposes exactly into corrected and corrupted predictions.

2.5 Statistics

Participants were the unit of inference. Task-effect intervals come from a participant-cluster bootstrap with 20,000 resamples, with seeds averaged within each resample rather than

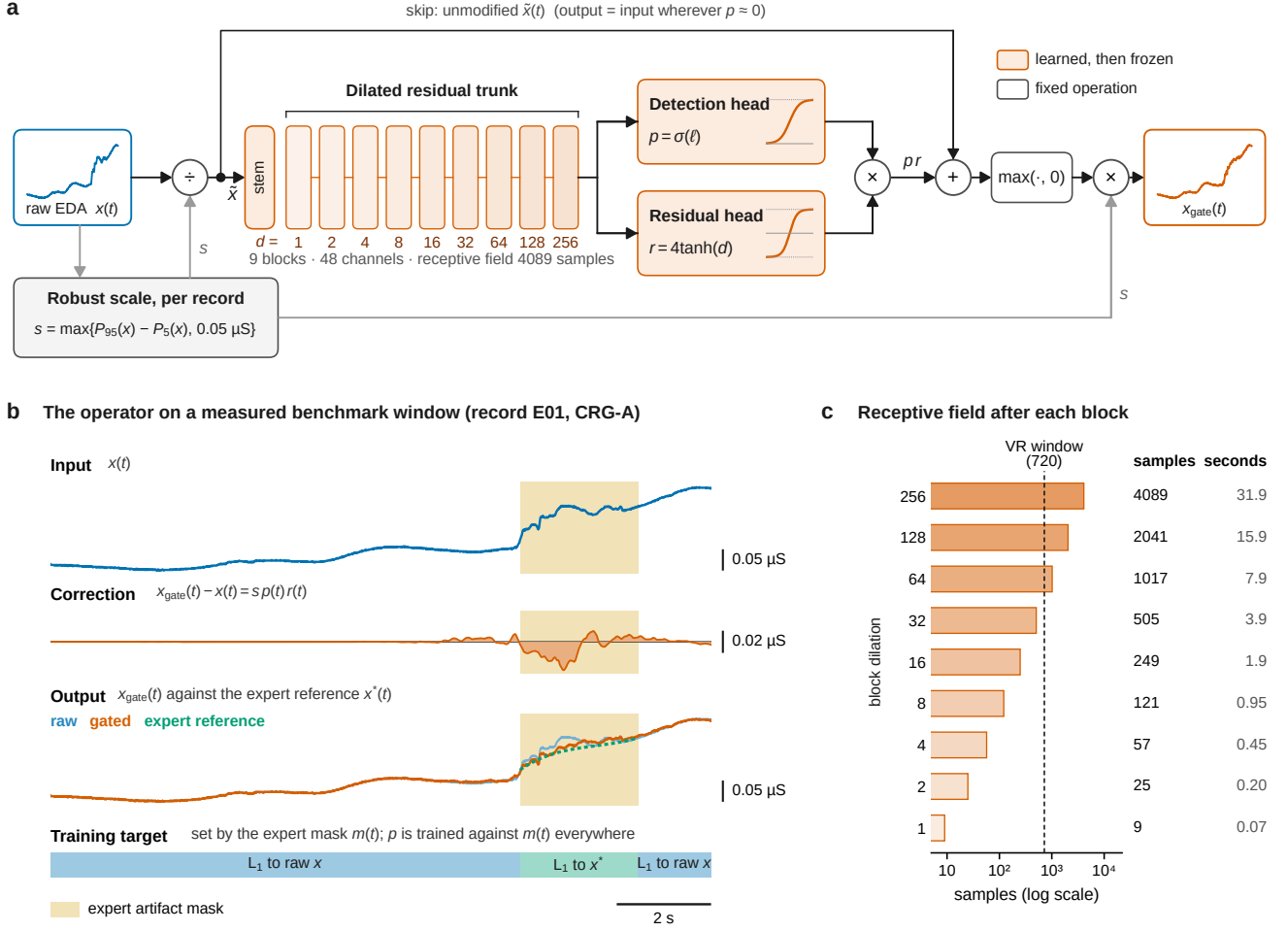


Figure 2: The continuous residual gate. (a) The raw trace $x(t)$ is divided by a per-record robust scale s . A stem and nine dilated residual blocks (48 channels, dilation d from 1 to 256) feed a detection head, whose output $p(t) = \sigma(\ell(t))$ lies between 0 and 1, and a residual head, whose output $r(t) = 4 \tanh(d(t))$ lies between -4 and 4 . Their product is added to the unmodified normalized trace, clamped at zero and multiplied by s (equations 1 and 2), so the output equals the input wherever $p(t)$ is close to zero. Shaded modules are learned on the benchmark and then frozen; the other operations have no parameters. (b) The operator on a 14-s window of benchmark record E01 (CRG-A output, 128 Hz, unfiltered). The correction $x_{\text{gate}} - x = s p(t) r(t)$ is concentrated in the expert-annotated artifact span, where the output moves towards the expert reference $x^*(t)$. The bottom track shows the training targets that the mask defines: an L_1 loss to x^* inside the span and to the raw signal outside it. The detection head is trained against the mask throughout, and CRG-B adds distillation from a fold-local teacher. (c) Receptive field after each block, from 9 samples after the first block to 4089 samples (31.9 s at 128 Hz) after the last. The dashed line marks the length of one VR window (720 samples).

treated as independent observations. Equivalence was tested with two one-sided tests (TOST) on seed-averaged participant differences against a margin of ± 3.32 pp, the benefit that oracle repair produced in the injection experiment (section 3.6). This margin was benchmarked on our own data rather than preregistered, and we also report the tightest margin at which all five families were equivalent. Bayes factors for a point null used a JZS prior with Cauchy scale $r = 0.707$ against a two-sided alternative [23]. Design sensitivity is given as the minimum detectable effect for 80% power at $\alpha = 0.05$, $\text{MDE} = 2.80 \times \text{SE}$, where SE is the bootstrap standard error; the mean across families was 1.44 pp. Paired benchmark comparisons used Wilcoxon signed-rank tests across records.

For the analysis of participant composition, each VR window received a label-blind proxy for contamination: the mean absolute gated-minus-raw change divided by the record scale. Raw-input accuracy, averaged over models and seeds, was compared between low- and high-proxy windows after a median split over all windows and after a median split within each participant. The within-participant contrast weights participants equally, and bootstrap resamples kept repeated participants as distinct clusters.

2.6 Controlled injection and contamination estimate

To estimate how much benefit repair could provide in this task, we added known artifacts to the VR windows. Artifact residuals, $a(t) = x_{\text{raw}}(t) - x_{\text{clean}}(t)$, were taken from expert-marked regions of the benchmark, resampled to 58.8 Hz and placed without reference to the labels:

$$x_{\text{cont}}(t) = \max\{x_{\text{vr}}(t) + s_{\text{vr}} \hat{e}(t), 0\}, \quad (3)$$

where $\hat{e}(t)$ is the placed residual in units of the record scale s_{vr} . Dose is expressed as the effective perturbed fraction, the share of samples whose injected magnitude exceeds 0.01 of the record scale: 4.42%, 8.94% and 16.70% for nominal doses of 5%, 20% and 40%. At each dose, four arms were compared with the contaminated input. Oracle localization interpolated across injected samples using their known positions but not their values. The gate applied its learned correction. Matched noise added a phase-randomized surrogate with the same RMS and power spectrum as the gate's correction, which tests whether any change of that size helps. The clean-input arm used the native windows. Within each dose, fold and seed, MiniRocket fitted its transform on contaminated training data and shared it across arms; because the transform was refitted at each dose, dose comparisons describe the adaptive pipeline rather than the signal alone. The prespecified validity check

was a loss for catch22 + RF at the highest dose relative to native input.

Natural contamination in the VR recordings was estimated with the transferred detector at a threshold of 0.930, where benchmark precision is 0.800 and recall 0.435. The flagged fraction was multiplied by precision and divided by recall. As a sensitivity analysis we held sensitivity and specificity fixed instead.

2.7 Literature sample

To gauge how often applied studies test preprocessing against the task, we searched PubMed, Europe PMC, Crossref and OpenAlex for studies published from 2019 to 2026 that recorded human EDA, handled artifacts and reported downstream inference [24–34]. Of 363 candidates, 108 passed title screening and 80 were screened in full, leaving 13 eligible studies. A study counted as making the comparison if it evaluated processed and unprocessed signals with the same task, model and protocol; comparisons between processing methods, or signal-quality scores alone, did not count. Screening used titles only rather than titles and abstracts as planned, coding was not independently duplicated, and Scopus and Web of Science were not available. We therefore treat the 13 studies as an illustrative sample rather than an estimate of prevalence. One study was coded from its abstract alone and counted as not making the comparison; counting it the other way would raise the proportion from 2/13 to 3/13.

3 Results

3.1 Gating did not improve classification

Gating did not improve classification in any model family (figure 3a). Changes in balanced accuracy ranged from -1.35 pp for catch22 + RF to $+0.93$ pp for MiniRocket. No family had an interval wholly above zero, and two had intervals wholly below it: catch22 + RF (-1.35 pp, 95% CI -2.38 to -0.25) and HYDRA (-1.00 pp, -1.98 to -0.05). The most favourable upper bound was $+2.05$ pp, for MiniRocket. In every family, individual participants fell on both sides of zero.

The data also bound the effect. All five families were equivalent to raw input within the tested margin of ± 3.32 pp (all $p \leq 0.003$), and the tightest margin at which all five were equivalent was about ± 2.37 pp (figure 3b). With a mean design MDE of 1.44 pp, effects large enough to justify a preprocessing step were within reach of the design. Bayes factors did not support an exact null: the largest BF_{01} was 2.88 (MultiRocket), and catch22 + RF gave weak evidence for a nonzero effect ($\text{BF}_{10} = 2.08$). Gating therefore changed accuracy by less than two to three points, with a tendency toward harm in two families and no sign of benefit in any. The rest of this section asks why a gate that performs well on its own benchmark delivered nothing here (table 2).

3.2 The gate repaired waveforms on the benchmark

Figure 4 shows two measured excerpts chosen to illustrate failure modes of CRG-A: the correction departs from the reference inside annotated spans at high artifact burden, and it intervenes where no artifact was annotated at low burden. Across all records, however, the detector ranked artifact samples well. The median record-level AUROC was 0.9409 (0.9477 in training records and 0.9263 in test records), and 36 of 43 records exceeded 0.90 (figure 5a). Its probabilities were too high, however. In every record the mean predicted probability exceeded the annotated artifact fraction (figure 5c), and the median mean probability (0.209) was 3.13 times the median artifact fraction (0.067). The reliability curve lies well below the identity line over most of the probability range (figure 5b). The detector thus ranks artifacts correctly but intervenes more often than the annotation warrants.

Reconstruction improved on average. Across the 43 records, artifact-region mean absolute error fell from 0.1010 to $0.0830 \mu\text{S}$, a reduction of 17.8% (paired Wilcoxon $p = 1.2 \times 10^{-4}$; figure 6a). The improvement was uneven. It came almost entirely from records annotated by Expert 1, where error fell by 27.6% and decreased in all 21 records, whereas error in Expert 2’s records was unchanged on average and fell in only 10 of 22 (table 3).

3.3 Lower waveform error came with worse event detection

The same CRG-B correction that lowered waveform error made SCR detection worse. Peak F_1 fell from 0.8980 to 0.8674 (-3.05 pp; $p = 2.3 \times 10^{-13}$), and it fell in all 43 records, in both expert subsets and in both splits (figure 6b, table 3). Figure 6c shows the two outcomes together: 31 records gained waveform accuracy while losing event accuracy, and the other 12 lost both. The gated signals also contained more peaks. The total excess of detected over reference peaks rose from $11,108$ to $12,989$ ($+16.9\%$), and the median ratio of detected to reference peak counts rose from 1.078 to 1.086 . A correction can therefore bring a trace closer to the reference on average while adding small deflections that a peak detector reads as responses. Waveform error alone does not show whether events are preserved.

3.4 Feature changes were widespread and weakly aligned

Classifiers do not detect SCRs explicitly, so the next question is how the correction moved the features they use. For a frozen feature map ϕ , we compared the displacement that the expert reference would require with the displacement the gate produced:

$$\Delta_{\text{oracle}} = \phi(x_{\text{clean}}) - \phi(x_{\text{raw}}), \quad \Delta_{\text{gate}} = \phi(x_{\text{gate}}) - \phi(x_{\text{raw}}). \quad (4)$$

In 57.4% of benchmark windows the reference required no change in MiniRocket features, yet the gate (CRG-A) changed the features of every one of these windows. Where change was needed, the gate moved features only partly in the right direction: the median cosine between the two displacements was 0.363 (68.7°), and the gate’s displacement was about half the required size (median ratio 0.507). The gate therefore changed features where nothing needed changing and corrected them only partly where something did. Comparable summaries for catch22 are not reported because their normalization could not be reconstructed from the archive.

3.5 Most decisions did not change

Feature movement matters only if it changes a prediction. Each paired prediction either keeps its class, becomes correct or becomes incorrect, so the change in pooled accuracy is exactly

$$\Delta \text{ACC} = \frac{N_{\text{corrected}} - N_{\text{corrected}}}{N}. \quad (5)$$

Averaged across families, 92.83% of the $2,952$ predictions per family (984 windows $\times 3$ seeds) kept their class, 3.46% became correct and 3.71% became incorrect (figure 7). The changes that did occur nearly cancelled. Families that changed more predictions tended to lose more accuracy (Spearman $\rho = -0.90$ across the five families, exact $p = 0.083$). Five related families are too few to establish this, but the direction fits a correction whose useful and harmful effects are of similar size.

3.6 Little benefit was available to recover

The last explanation concerns how much contamination there was to remove. Corrected for benchmark precision and recall, the transferred detector implies an artifact burden of 4.64% in the VR recordings; holding sensitivity and specificity fixed

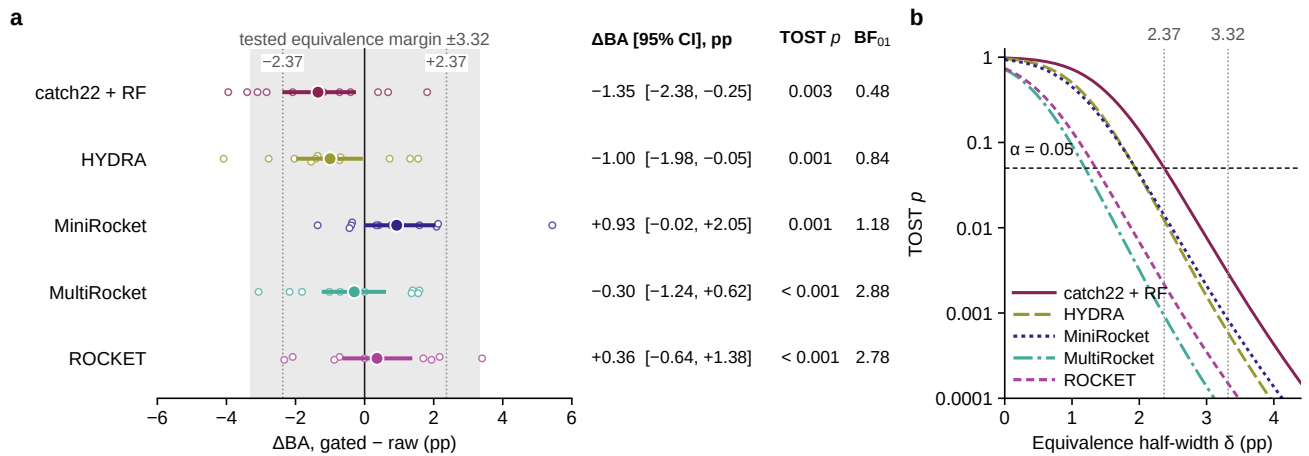


Figure 3: Effect of gating on VR classification. (a) Gated-minus-raw balanced accuracy for each model family. Open circles are the 11 held-out participants, each averaged over seeds; filled circles and bars are the mean and its 95% participant-cluster bootstrap interval. The shaded band is the ± 3.32 pp margin tested by TOST, and dotted lines mark the tightest common margin (± 2.37 pp). BF_{01} uses a JZS prior ($r = 0.707$) against a two-sided alternative. (b) TOST p as a function of the equivalence half-width δ , computed from the same participant differences (paired t , 10 degrees of freedom). A family is equivalent within $\pm \delta$ wherever its curve lies below $\alpha = 0.05$.

Table 2: From contamination to decision: where the expected benefit was lost. Each row is a separate measurement; the rows describe one method, not one output traced through every stage.

Stage	Measurement (checkpoint)	Result	Meaning
Contamination	Artifact share	10.75% annotated in the benchmark; 4.64% estimated in VR	Less to remove in VR
Detection	Record AUROC (CRG-A)	Median 0.9409; 36/43 records above 0.90; median probability 3.13 times median artifact fraction	Good ranking, too much intervention
Reconstruction	Artifact-region MAE (CRG-B)	0.1010 to 0.0830 μ S (-17.8%)	Waveforms improved
Events	SCR peak F_1 (CRG-B)	0.8980 to 0.8674; lower in 43/43 records	Events degraded
Features	MiniRocket displacement (CRG-A)	No change needed in 57.4% of windows, all changed; median cosine 0.363	Change where none was needed, partial alignment where it was
Decisions	Prediction classes, VR (CRG-B)	92.83% unchanged; 3.46% corrected; 3.71% corrupted	Changes nearly cancel
Task	ΔBA, VR (CRG-B)	-1.35 to +0.93 pp; equivalent within ± 3.32 pp	No benefit
Available benefit	Injection, catch22 + RF (CRG-A, window-wise)	Oracle +3.32 pp; extrapolated to native burden 1.05 pp	Little to recover

Table 3: CRG-B reconstruction and event outcomes by expert subset and split. MAE is averaged across records; reduction = $100 \times (1 - MAE_{\text{gated}}/MAE_{\text{raw}})$. The last two columns count records. Expert and split subsets overlap, so rows are not additive. In the test split alone, F_1 fell by 2.50 pp ($p = 0.002$) and MAE did not change significantly ($p = 0.084$; paired Wilcoxon).

Subset	n	MAE (μ S) raw $\rightarrow$ gated	Red. (%)	MAE lower	F_1 lower
Expert 1	21	0.134 $\rightarrow$ 0.097	27.6	21/21	21/21
Expert 2	22	0.069 $\rightarrow$ 0.070	-0.2	10/22	22/22
Training	33	0.102 $\rightarrow$ 0.085	16.3	25/33	33/33
Test	10	0.098 $\rightarrow$ 0.075	23.1	6/10	10/10

instead gives 2.88%. Both estimates assume that detector performance transfers from the benchmark, which cannot be checked without expert labels in VR, but both are well below the 10.75% annotated in the benchmark.

The injection experiment measured what repair could achieve when contamination is known (figure 8). For catch22 + RF, oracle localization recovered 3.32 pp relative to contaminated input, pooled over the three nonzero doses (95% CI 1.86 to 4.76), whereas matched noise cost 2.11 pp (-3.82 to -0.08). The gate recovered 1.16 pp (-1.13 to 3.31), about a third of the oracle's benefit. For MiniRocket, oracle localization had no effect (-0.29 pp, -1.81 to 1.06). Repair can help a feature-based classifier when contamination is substantial and

correctly located, but not every classifier is sensitive to it.

The prespecified validity check did not reach significance. At the highest dose, catch22 + RF lost 4.13 pp relative to native input (-8.33 to +0.57). The oracle and matched-noise contrasts show that the experiment could detect both benefit and harm, but they do not turn the failed check into a pass. Extrapolating the injection slopes to the estimated native burden gives expected benefits of 1.05 pp for catch22 + RF and 0.23 pp for MiniRocket, with slope intervals that include zero (Supplementary Note N7-4). At the contamination present in this task, the benefit available even to perfect repair is small and below the mean design MDE of 1.44 pp.

3.7 Participant composition can make cleaning look useful

A pooled association between artifact level and accuracy can make cleaning look necessary before any processed-versus-raw comparison is run. In our data, windows with low proxy values were classified 11.25 pp more accurately than windows with high values after a global median split (95% CI 1.52 to 17.31). When the split was made within each participant, the difference was 0.07 pp (-3.97 to 4.10; figure 9a). The pooled gap arose because high-proxy windows were concentrated in a few participants. The share of a participant's windows above the global median ranged from 9.2% to 95.3%, and three participants supplied 50.4% of all high-proxy windows (figure 9b). Participants who are harder to classify can therefore

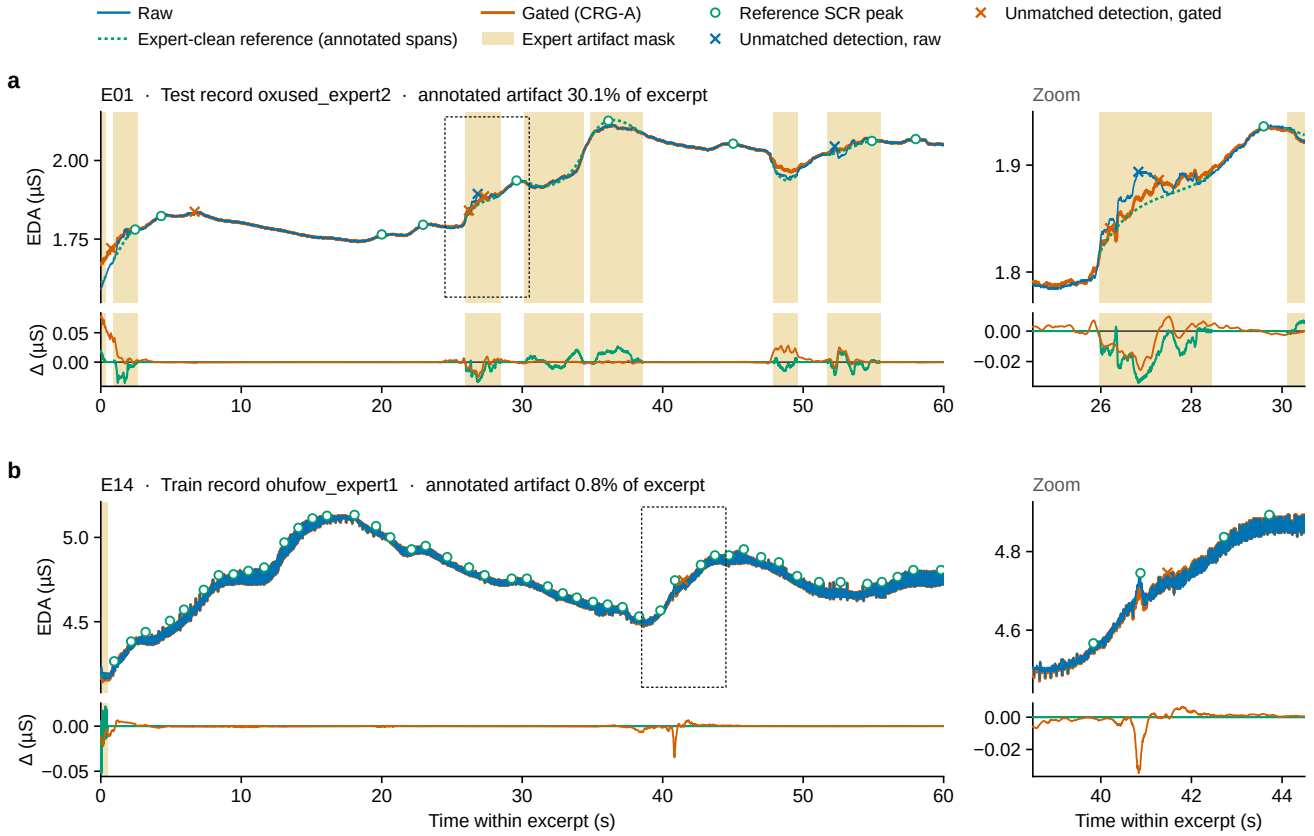


Figure 4: Two measured benchmark excerpts at 128 Hz, unfiltered, selected to illustrate failure modes rather than typical behaviour: gate failure at high artifact burden in E01 (test record) and over-intervention at low burden in E14 (training record). Upper panels show the raw trace (blue) and the CRG-A output (vermillion), which is drawn beneath raw so that it is visible only where the gate moved the signal. The expert-clean reference (green, dotted) is drawn within annotated spans; outside them it equals raw in E14 and differs from raw by less than $0.0025 \mu\text{S}$ in E01. Lower strips show reference minus raw (green) and gated minus raw (vermillion). Circles are reference SCR peaks, and crosses are detections on the raw or gated trace left unmatched by one-to-one matching within ± 1 s. Dashed boxes mark the windows enlarged at right.

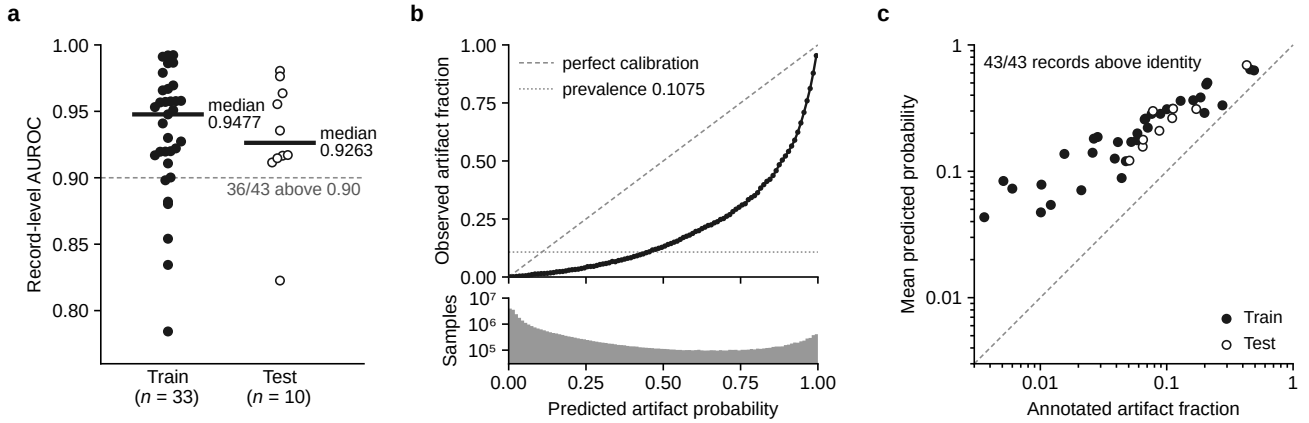


Figure 5: Detector behaviour on the benchmark (CRG-A; out-of-fold predictions for training records, test-ensemble predictions for test records). (a) Point-wise AUROC for each record; bars mark split medians, and 36 of 43 records exceed 0.90. (b) Observed artifact fraction in 100 fixed probability bins of width 0.01, with the number of samples per bin below on a log scale (34,312,745 samples in total). (c) Mean predicted probability against annotated artifact fraction for each record. All 43 records lie above the identity line.

create an apparent link between artifacts and accuracy. Because the thresholds and weighting also change between the two analyses, the attenuation does not measure the participant effect exactly, but it is enough to show that the pooled association is not evidence that cleaning helps.

3.8 Sparser intervention did not preserve events

The continuous gate changed almost every sample (99.8%), which raises the question of whether a sparser rule would do better. A binary variant of the same student, which corrects only samples classified as artifact, changed 25.0% of samples. It did not differ from the continuous gate in waveform error

($p = 0.372$) or SCR F_1 ($p = 0.071$). Both gates reduced F_1 relative to raw input, and the binary gate produced a higher ratio of detected to reference peaks ($p < 10^{-4}$). Sparsity alone therefore did not preserve events. A more useful design target is the smallest correction that meets a fidelity requirement, checked against event and task outcomes rather than waveform error alone.

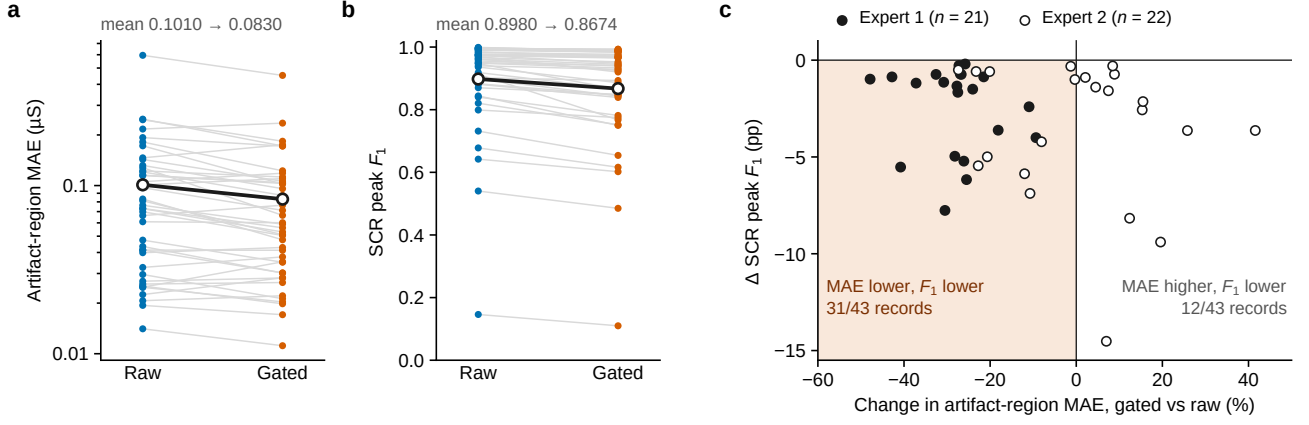


Figure 6: Waveform repair and event preservation on the 43 benchmark records, CRG-B continuous-strength student versus raw input. (a) Artifact-region mean absolute error to the expert-clean reference, on a log scale. (b) SCR peak F_1 . Thin lines join each record and the thick line joins the means. (c) The same records as paired changes. Waveform error fell in 31 records and event F_1 fell in all 43. Filled circles are records annotated by Expert 1 and open circles those annotated by Expert 2.

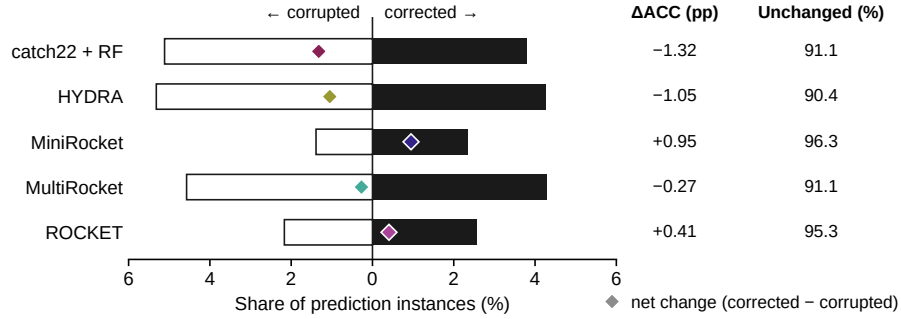


Figure 7: Decision changes after gating for each model family, with 2,952 prediction instances per family (984 windows $\times$ 3 seeds). Bars to the right count instances that became correct, and bars to the left those that became incorrect. Diamonds mark the net change, which equals the pooled accuracy difference in equation 5. The right-hand columns give that difference and the share of instances whose class did not change.

4 Discussion

4.1 Why a better waveform did not give a better decision

Artifact removal is usually justified by the quality of the cleaned waveform. In this study the waveform improved, yet the decision built on it did not. The benchmark measurements account for most of the gap. The correction that lowered waveform error added small deflections that degraded event detection in every record. It changed features in windows that needed no change and corrected them only partly in windows that did. Its effect on decisions was small and two-sided, so corrected and corrupted predictions nearly cancelled. The VR recordings also contained little contamination, so even perfect repair had little to recover. These explanations act together, and our design does not divide the null result among them, but they lead to the same conclusion: waveform fidelity is a poor guide to the value of preprocessing in a downstream task.

One explanation cannot be excluded. The VR corpus has no expert reference, so we could not measure how well the gate repaired the VR signals themselves, and transfer from 128 Hz benchmark records to 58.8 Hz VR recordings may have weakened the correction. This does not change the practical conclusion. Anyone who applies a validated artifact handler to new data faces the same uncertainty, and only the processed-versus-raw comparison answers the question for their task.

4.2 Relation to prior work

Our findings agree with a growing body of work showing that preprocessing which improves a signal by its own criteria need not improve the analysis that follows. In EDA, analytic choices

change SCR estimates [6, 7, 11], and filtering and model inversion change sensitivity and predictive validity [8–10]. In EEG, Delorme [13] found that, apart from high-pass filtering and bad-channel interpolation, automated corrections left the number of channels with significant condition effects unchanged or reduced it, and Kang et al. [14] found that component-based artifact rejection did not improve deep-network decoding in three brain-computer interface tasks. Reconstruction methods for EDA, from wavelet denoising [35] to learned models [16], are evaluated against clean waveforms. That evaluation is necessary but, as our results show, not sufficient. This study adds a paired downstream comparison combined with expert-referenced measurements of where the expected benefit is lost, from waveform to events, features and decisions, in a VR classification task.

4.3 Implications for practice

Three practices follow. Any EDA pipeline that removes artifacts should keep an unprocessed arm and compare the two on the intended decision, using identical windows, participant-disjoint folds, transforms and classifiers fitted on training participants only, and participant-level uncertainty; negative results from this comparison should be reported. A disappointing comparison should then be diagnosed with event preservation, the extent of intervention and the balance of corrected and corrupted decisions, not with waveform error alone. Finally, associations between artifact level and accuracy should be checked within participants before they are used to argue for cleaning. For the design of artifact handlers, our results favour the smallest correction that meets a fidelity

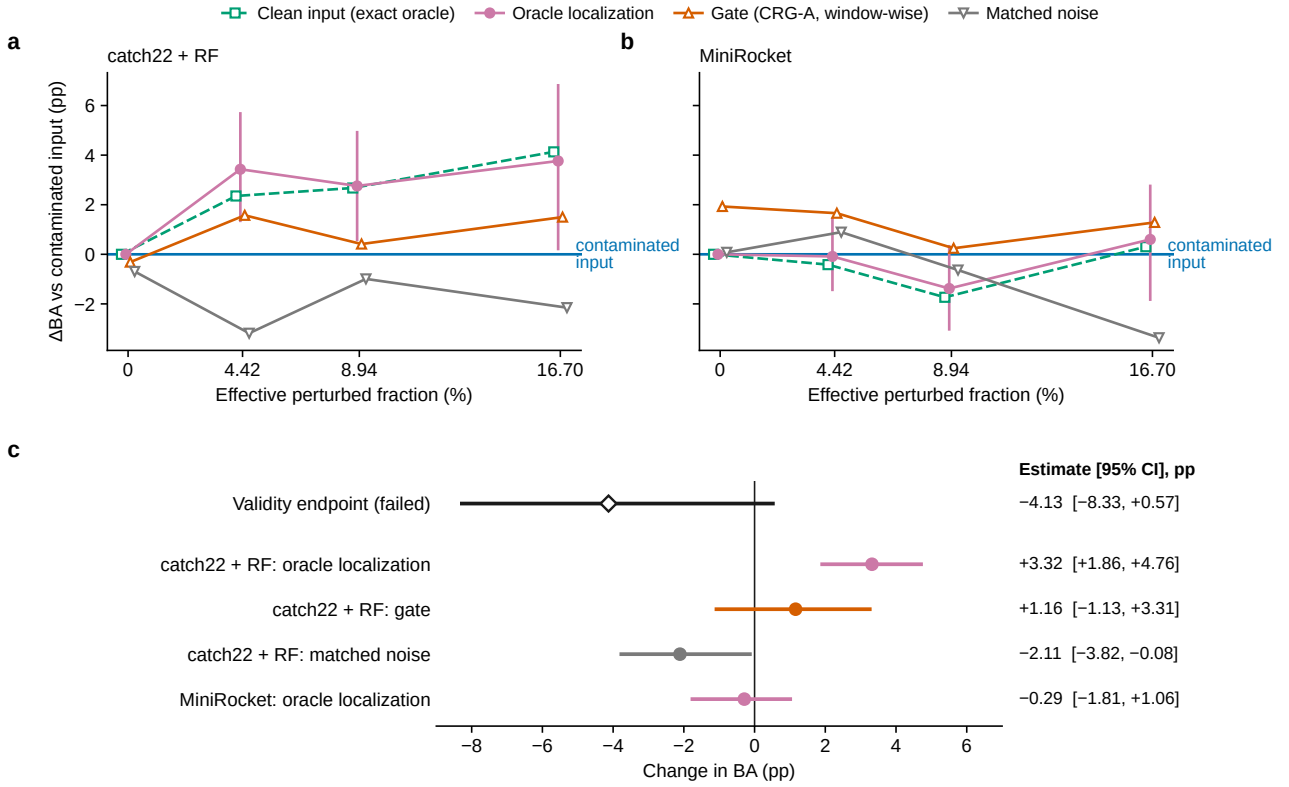


Figure 8: Controlled artifact injection. (a, b) Change in balanced accuracy relative to contaminated input at each effective perturbed fraction for catch22 + RF and MiniRocket; contaminated input defines the zero line. Points are means over 11 participants with seeds averaged, and arms are offset horizontally for legibility. Vertical bars are 95% participant-cluster bootstrap intervals for the oracle-localization contrast. The clean-input arm reuses the native windows at every dose, so its curve shows the damage that perfect repair would undo; MiniRocket’s values vary across doses only because its transform is refitted at each dose. (c) The prespecified validity check (catch22 + RF, highest dose versus native input) and pooled nonzero-dose contrasts against contaminated input.

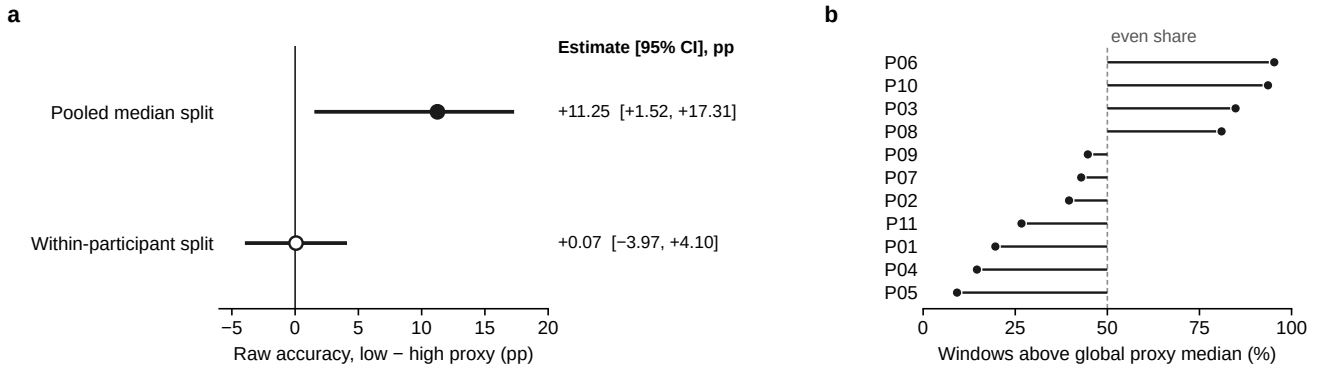


Figure 9: Participant composition behind the artifact-accuracy association. (a) Difference in raw-input accuracy between low- and high-proxy windows after a global median split and after a median split within each participant, with 95% participant-cluster bootstrap intervals. (b) Share of each participant’s windows above the global proxy median; without differences between participants every share would lie near 50%. Three participants (P06, P10 and P03) supply 50.4% of all high-proxy windows.

requirement over corrections that change most of the signal. Systems meant to warn of a loss of balance before it happens will also need tests of lead time and prospective reliability, which this post-onset task does not provide.

4.4 Limitations

Expert point-wise annotation exists for only one corpus, and every statement about ground truth rests on it. Its two expert subsets do not overlap, so inter-rater agreement could not be measured, and comparisons between the subsets (AUROC $p = 0.248$, artifact fraction $p = 0.202$, change in SCR F_1 $p = 0.474$) do not replace it. The downstream corpus has 11 participants, which limits power; Bayes factors were inconclusive, so we bound the effect rather than claim its absence. The VR burden estimate assumes that detector precision and

recall transfer from the benchmark, and it was computed on 120-s excerpts per participant rather than full recordings. An annotation-free plausibility score planned as a second estimate failed calibration (maximum precision 0.546 at 58.8 Hz against a target of 0.800), possibly because three quantities of different scales were summed without weighting. In the injection experiment the gate was applied to 720-sample windows, shorter than its receptive field of about 69.5 s at 58.8 Hz, so its absolute performance there is not comparable with the full-trace gate, although comparisons across doses remain valid; the prespecified validity check also failed. The decision analysis rests on five related model families and is exploratory. Different analyses used two checkpoints of the same architecture (section 2.3). Our results concern one task, one artifact handler and five classifier families, and tasks with heavier contamination may

benefit more from repair.

5 Conclusion

Artifact removal improved electrodermal waveforms on a benchmark with expert references but did not improve classification of user state in a VR balance-disturbance task. The expected benefit was lost because waveform repair degraded event detection, changed features where no change was needed, left most decisions unchanged, and had little contamination to remove. Preprocessing should be judged by the decision it supports, with an unprocessed arm as the comparison and a within-participant check before artifact-accuracy associations are taken as evidence for cleaning.

Data and code availability

Both corpora are public under a CC BY 4.0 licence: EDABE v2 (Mendeley Data, <https://doi.org/10.17632/w8fxrg4pv5.2>) and the VR balance-disturbance corpus (Zenodo, <https://doi.org/10.5281/zenodo.14013468>). The accompanying repository contains the frozen analysis outputs, the code that regenerates the included figures and the analysis scripts. It does not contain all upstream waveforms and injection arrays or the literature candidate-selection files. A persistent repository DOI will be added after archive registration.

Declaration of competing interest

[all-author competing-interest declarations pending].

CRedit author contribution statement

[author-specific CRediT roles pending].

Funding

[funders, grant numbers and sponsor roles pending].

Ethics and data use

This secondary analysis uses public data released under CC BY 4.0. [institutional ethics determination and supporting record pending].

Use of generative AI

Generative AI tools (OpenAI Codex and Anthropic Claude) assisted with drafting, language editing and figure code. [authors' confirmation that they reviewed and edited the output and take full responsibility for the content pending].

Acknowledgments

[acknowledgments and permission to name contributors pending].

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